

MA 211: Ordinary Differential Equations

Course Code	MA 211
Course Name	Ordinary Differential Equations
L-T-P-C	3-1-0-4
Prerequisites	IC113, IC115
Intended for	B. Tech 2nd Year (MnC) and optional for other branch of students
Mutual Exclusion	MA513 course from SMSS is not allowed to credit by the students after or along with this course
Approval	54th BoA (May 2024)

Course Contents

- **Module I:** General Overview of Ordinary Differential Equations; Solutions methods to solve the first, second and higher order ordinary differential equations; Power Series methods, with properties of Bessel differential equations and Legendre differential equations. (12 Hours)
- **Module II:** Existence and Uniqueness of solutions for Initial Value Problems: Picard's and Peano's Theorems, Gronwall's inequality, continuation of solutions and maximal interval of existence, continuous dependence. (14 Hours)
- **Module III:** Algebraic properties of systems of differential equations, the eigenvalue-eigenvector method of finding the solutions of linear systems, Complex eigenvalues, Equal eigenvalues, Fundamental matrix solutions, Wronskian, Matrix exponential, Nonhomogeneous equations, Variation of parameters, Stability theory for linear and nonlinear systems, Lyapunov stability. (16 Hours)

Text Books

1. Ahmad, S. Rao, M.R.M., **Theory of ordinary differential equations with applications in biology and engineering**, EWP publication,
2. L. Perko, **Differential Equations and Dynamical Systems**, Texts in Applied Mathematics, Vol. 7, 2nd Edition, Springer Verlag, 1998.

References

1. Devaney, R., Hirsch, M. W. and Smale, S., **Differential Equations, Dynamical Systems, and an Introduction to Chaos**, 2nd Edition, Academic Press, 2003.
2. Birkhoff, G. and Rota, G.-C., **Ordinary Differential Equations**, wiley, 1989
3. R. P. Agarwal and D. O. Regan, **An Introduction to Ordinary Differential Equations**, Springer- Verlag, 2008.